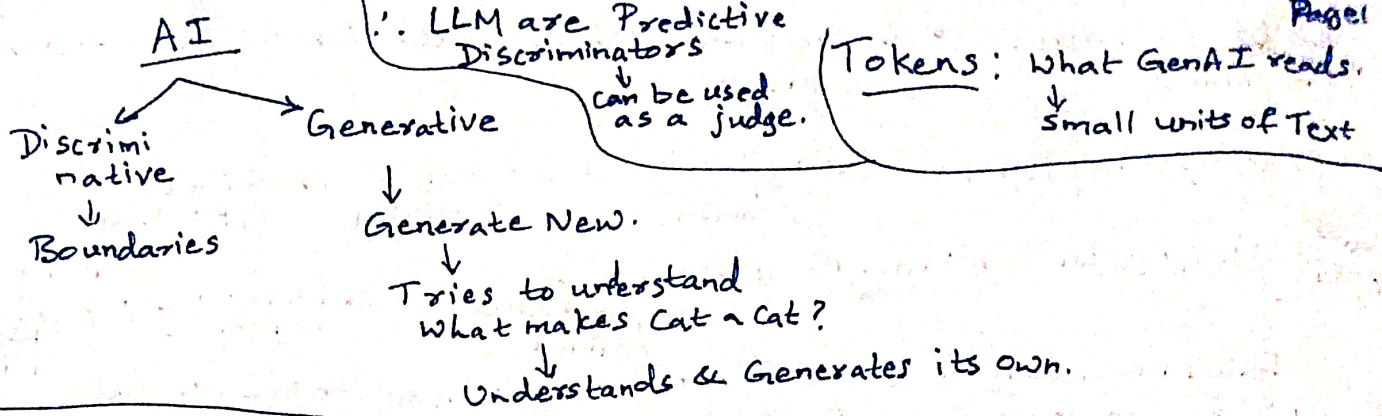
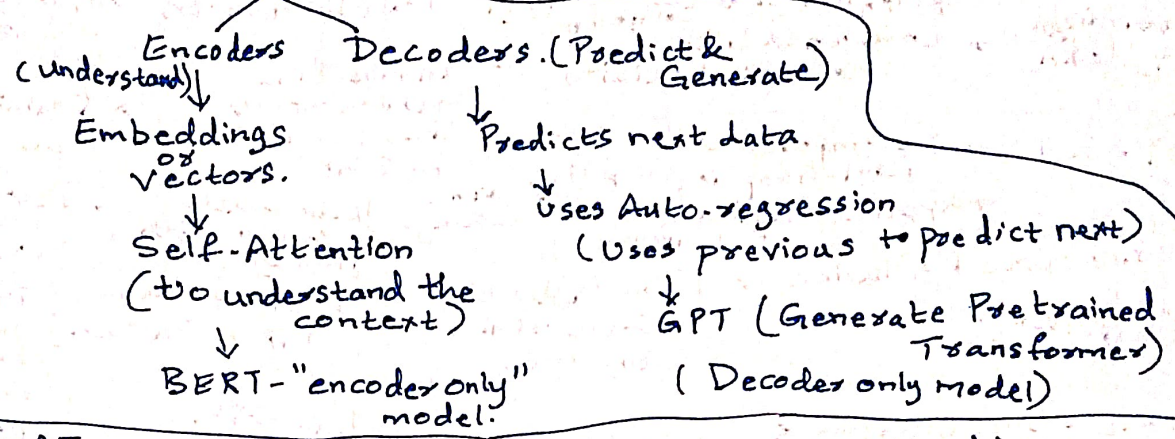
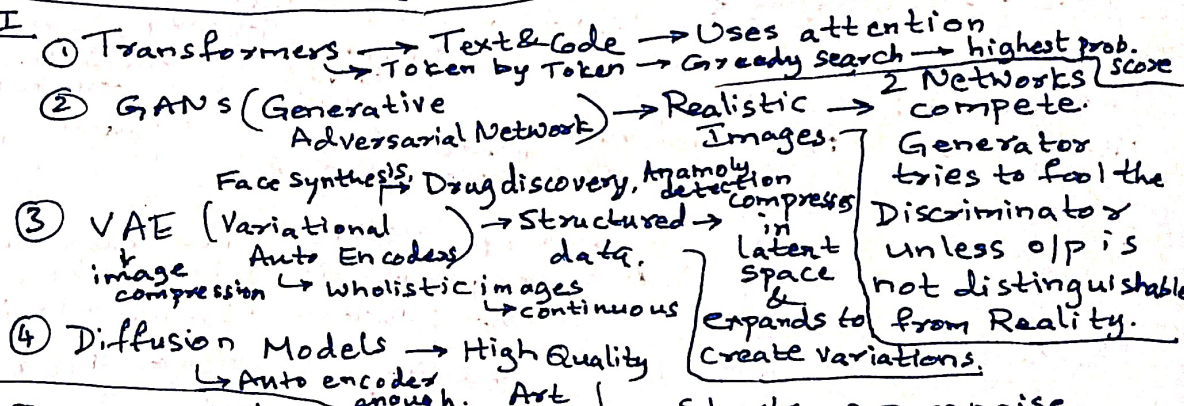




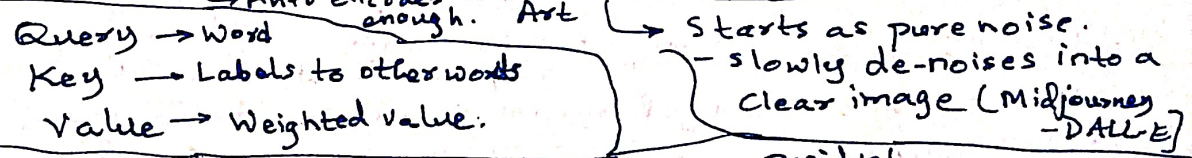
Modern GenAI → Transformers.



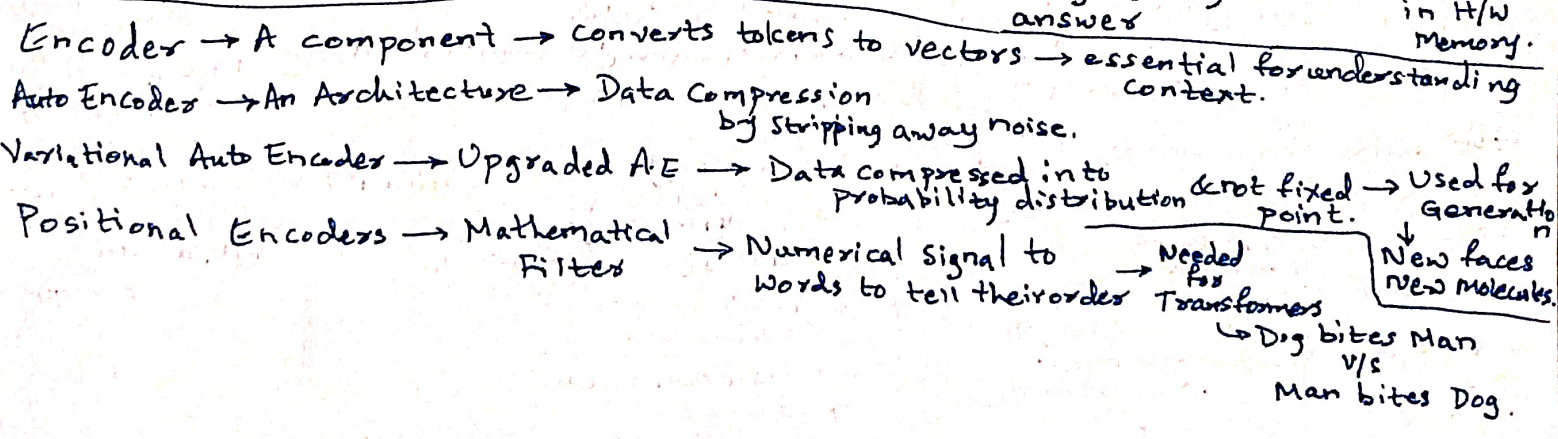
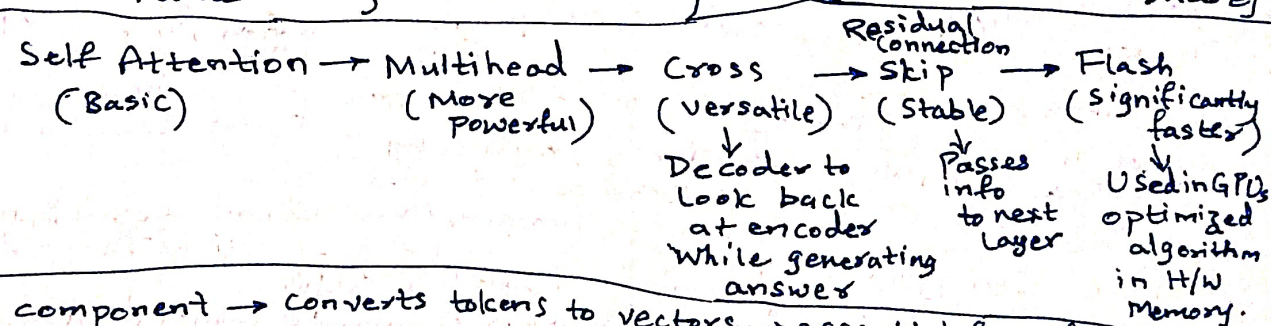
Models of GenAI



Self Attention



Evolution:



Techniques → A pre-trained Model updates its weight on new, smaller dataset.
Fine Tuning

Prompting / RAG → Lowest Level of Training → 0 Parameters → Minimal GPU Usage.

Called PEFT { LoRA / QLoRA → Mid Level Training → 0.1% to 3% of parameters → Can Run on Laptop. } → PEFT Parameter Efficient Fine Tuning

Full Fine Tuning → Highest → 100% of parameters → Massive - need GPU Data centers.

PEFT: → When you need Domain expertise on top of Foundational LLMs. → Legal, Medical etc.

LoRA → Low Rank Adaptation → Freeze Main Model
 ↳ Applies to direction component. Add Tiny Trainable Adapters → 98% Less Memory → Use Human Labeled Ground Truth Data.

Quantized PEFT LoRA → Compressed LoRA to 4 Bit Precision → can fine tune massive models like Llama (70B) in 1 GPU → Training slower due to Decompression

PEFT DoRA → Decomposed Low Rank Adaptation → Eg: Cannot change Radio Station & Volume with 1 knob. → Decomposes Weights Magnitude & Direction Part → Achieves Full Fine Tune Perf

Knowledge Distillation → Tiny Model that behaves similar to big one. → o/p - Student I/P → Teacher. → Instead of Learning from RAW data, learns from Teacher's LOGITS. Eg: Teacher says 90% Dog, 10% Wolf. Student: Dog & Wolf Related.

SML → Small Language Model → New Architecture → Built with Fewer Parameters To Run on EDGE H/W. Eg: Phi-3 (MS) Google Gemma 2B

Foundational ML Models → Pretrained on massive data | General Purpose | Adapt to Many Tasks | Traditional ML. : Task Specific | Requires Labeled data | No Generalization

Hallucinations: Why?
 (A) Data Gaps or Noise
 (B) Over optimization
 (C) Context Overload.

Prevention Hallucination Mechanisms:

- ① RAG → Architecture → Looks up trusted Docu. marks.
- ② Grounding → Prompt → Forcing Model to cite sources.
- ③ Chain of Thought (CoT) → Logic → Asking to think step by step
- ④ Temperature or Randomness → Filtered → Parameters → Temp 0.0 → Greedy Search
- ④ Self Reflection → Validation → 1 LLM generates other LLM Critiques → Training Smaller student Model
- ⑤ Knowledge distillation with Fact checking → Model → Training Smaller student Model
- ⑥ N shot Prompting → Prompt → 3-5 examples of correct ans
- ⑦ Verification → Hallucination - score cascades o/p (RAGs or Arize Phoenix)

Neural Network Types: Used For

MLP (Multilayered Perception) → Basic Logic → Tabular Data, Simple classification

CNN → Vision → Image Recognition, Video Analysis

RNN → Sequence or Memory → Time Series, Speech to Text.

Transformers → Context or Reasoning → LLMs / Translation

Generative Adversarial Network → GAN → Creation or Mimicry → Creating Hyper Realistic fake images / Faces

Problem With RNN: ① Vanishing or Exploding gradient. "Forgets" beginning of sentence.

② Sequential Processing ③ Lack of Global Context

Cosine Similarity V/S Euclidian Distance

Criteria:	C.S [Vector DB]	E.D [Approx Neighbours]
Focus	Dir/Direction Orientation	Magnitude / Dist
NLP suitability	Excellent (Ignores Length)	Poor (sensitive to word freq & length)
Common use case	Semantic Search or Doc matching	clustering / Fraud Detection
Range	-1 to +1	Spatial data
Intuition	Are we thinking same thing?	How far apart are these points?

⑦ Verification → Hallucination - score cascades o/p (RAGs or Arize Phoenix)